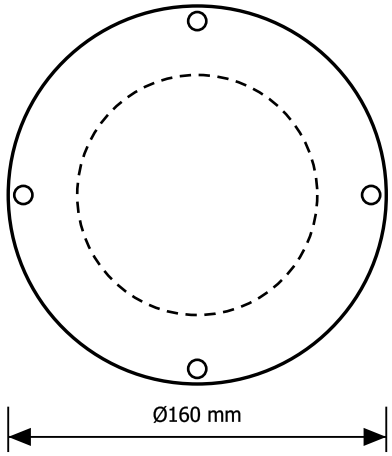
HACKERFAB IITB | SEMICONDUCTOR HOT PLATE

PRELIMINARY CONCEPT DRAWING - NOT FOR MANUFACTURE

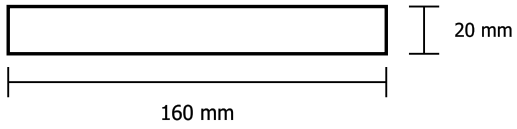
TOP VIEW



Dashed circle: wafer reference Ø101.6

Four heater positions are schematic

FRONT / EDGE VIEW



Schematic side profile; verify detailed design

DESIGN BASIS AND REQUIREMENTS

Plate: nominal Ø160 x 20 mm; wafer reference Ø101.6 mm (4 inch).

Heating: 4 x nominal 250 W cartridge heaters, concept Ø6 x 60 mm.

Sensing: 2 x four-wire Pt100 RTDs; locations need validation.

Operating target: ambient to 400 °C.

Setpoint error target: less than 5 °C.

Run-to-run difference target: less than 0.5 °C.

Uniformity target: ±0.5 °C in qualified area; max spread 1 °C.

Top flatness target: ±10 µm; verify by suitable metrology.

BILL OF MATERIALS - CONCEPTUAL

1 x aluminium plate; alloy/temper must be qualified at 400 °C.

4 x cartridge heaters; 2 x four-wire Pt100 sensors.

PID controller; rated SSR; contactor; fuse or circuit breaker.

Independent over-temperature cutoff; rated wires and terminals.

Thermal insulation; grounded enclosure; supports and fasteners.

ASSEMBLY AND SAFETY NOTES

1. Review alloy, heater fit, bore layout and high-temperature strength.
2. Fit heaters and RTDs; provide strain relief and thermal insulation.
3. Bond protective earth; use rated switching and independent cutoff.
4. Guard live terminals; verify wiring, insulation and grounding.
5. Map temperatures at multiple points; repeat cycles; measure flatness.

QUALIFICATION LIMITATIONS

Targets are not demonstrated until tested. Aluminium 6061 at 400 °C

requires review for temper loss, warpage and flatness retention.

Concept-level aid only; not a released manufacturing drawing.